Homework 07

REAL VARIABLES I

Fall 2026

Due: 2026-10-11 19:00
Topics: The Lebesgue dominated convergence theorem, approximations of the Dirac delta, convergence of integrals over subsets, and equi-uniform integrability with Vitali's convergence theorem

Instructions

Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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1. (Rescaling and the Dirac delta)

1.a

weight: 1.0

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous bounded function on $\mathbb{R}$. Prove that

$$\frac{1}{\pi} \lim_{n\to\infty} \int_{\mathbb{R}} f(x) \frac{n}{1+n^2x^2} \, \mathrm{d}\mathcal{L}(x) = Cf(0)$$

where $C = \int_{\mathbb{R}} \frac{1}{1+x^2} \, \mathrm{d}\mathcal{L}(x)$.

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Change variables $y = nx$.

- $\int_{-\infty}^{+\infty} \frac{1}{1+y^2}\,\mathrm{d}y = \pi$.
- Do not forget to justify any limiting procedure.

2. (Lebesgue dominated convergence theorem)

2.a

weight: 0.3

$$\lim_n \int_0^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.b

weight: 0.3

$$\lim_n \int_0^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.c

weight: 0.7 IN CLASS Find

$$\lim_{n\to\infty} \int_0^n \left(1-\frac{x}{n}\right)^n \mathrm{d}x.$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.d

weight: 1

3. (Integrals over subsets)

Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f_n : X \to [0, +\infty]$ be measurable and pointwise convergent to a function f. Assume

$$\int_X f_n(x)\,\mathrm{d}x \to \int_X f(x)\,\mathrm{d}x < \infty.$$

Prove that for any $E \subset X$

$$\int_E f_n(x)\,\mathrm{d}x \to \int_E f(x)\,\mathrm{d}x < \infty.$$

4. (Equiuniform integrability and Vitali's theorem)

Let $(X, \mathcal{A}, \mu)$ be a finite measure space, and let $f_n \colon X \to \mathbb{R}$ be a sequence of measurable functions that converges pointwise to a function f.

4.a

weight: 1 **IN CLASS**

Assume that

$$\sup_{n\in\mathbb{N}} \int_X |f_n(x)| \log(1 + |f_n(x)|)\ \mathrm{d}\mu(x) < +\infty$$

. Prove that

$$\lim_n \int_X f_n(x)\,\mathrm{d}\mu(x) = \lim_n \int_X f(x)\,\mathrm{d}\mu(x)$$

> **Free hint.** Prove that the sequence f_n is equi-uniformly integrable. Apply Vitali's convergence theorem.